Normalized Nonlinear Comparison on Dependency-Safe Local Cylinders

Abstract

We replace a high-dimensional global chart ball by an intersection of local four-link cylinders. A normalized restriction theorem converts local oscillation and reference-complement estimates into an explicit density comparison. At the frozen centered $SU(2)$ reference the loss is $3.02 \cdot 10^{-5}$ per support unit, closing that finite normalization gate. Uniformity over retained nonlinear boundary states remains open.

1 Restriction theorem

Let ν_s be a reference probability law, G_s a measurable good set, and

$$\mu_{s,G}(A) = \frac{\int_{A \cap G_s} e^{-R_s}\, d\nu_s}{\int_{G_s} e^{-R_s}\, d\nu_s}.$$

Theorem 1. *Suppose, uniformly in the retained state s,*

$$\nu_s(G_s) \geq 1 - \delta, \qquad \operatorname{osc}_{G_s} R_s \leq m\kappa, \qquad 0 \leq \delta < 1.$$

Then every measurable A satisfies

$$\mu_{s,G}(A) \leq \frac{e^{m\kappa}}{1-\delta}\, \nu_s(A).$$

The logarithmic comparison loss per support unit is

$$\kappa - \frac{1}{m} \log(1-\delta).$$

Proof. On G_s, divide the numerator and denominator by $e^{-\inf_{G_s} R_s}$. The numerator is at most $\nu_s(A \cap G_s)$, while the denominator is at least $e^{-\operatorname{osc}_{G_s} R_s} \nu_s(G_s)$. Insert the two hypotheses. □

If R_s is a sum of m local terms, each based on a radius-r cylinder, with vanishing value and gradient at its center and Hessian remainder at most H, then one may take $\kappa = Hr^2$.

2 Frozen-reference calibration

Stages 21 and 22 provide

$$m = 216, \quad r = 0.006023157807602598, \quad \delta \leq 0.002792708080357906, \quad H = 0.4746710865657404.$$

Consequently,

$$Hr^2 = 1.72203218 \cdot 10^{-5}, \qquad \frac{mHr^2 - \log(1-\delta)}{m} = 3.01676133 \cdot 10^{-5}.$$

This is far below the declared target 0.2. The global 408-dimensional ball used previously was therefore a representation mismatch: the relevant coverage event is local and overlapping, not a small Euclidean ball in every physical coordinate simultaneously.

3 Frontier

The theorem is uniform whenever its two hypotheses are uniform. The present calibration establishes them for the frozen centered reference, but not for every positive-mass retained nonlinear $SU(2)$ state. Nor does a restricted good-channel comparison control the bad partition. Those two quantifiers remain the physical frontier; no full-measure, RG, continuum, or mass-gap claim follows.